

Will Eagle - Curriculum Vitae

Jackson School of Geosciences

University of Texas at Austin

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<https://sites.google.com/utexas.edu/will-eagle?usp=sharing>

Education

B.S. General Geology, Jackson School of Geosciences, The University of Texas at Austin, May 2026. Thesis Advisor: Prof. Jay Banner.

Geologic Field Camp, Judson Mead Field Station, Indiana University Bloomington, July 2024.

Research Interests

I am a geoscientist interested in issues of karst water resources, and hydrogeology. I am currently focused on using drought signals preserved in tree rings to reconstruct past climate, and better inform water resource management. I am an avid caver, and conservation of karst resources is part of what motivates my work.

Current Research Projects

Surface Water Groundwater

Karst Hydrogeology Modulates Bald Cypress Tree-ring Climate Sensitivity: My undergraduate thesis focused on using tree rings in contrasting hydrogeologic settings to evaluate groundwater flow paths and optimize paleo-drought reconstructions. I found that perennially spring-fed trees underestimated droughts, and ephemeral spring-fed trees underestimated wet periods.

Previous Research Projects

Geochemical Evolution of Water in an Arid Cave: Geochemical modeling to constrain the extent of physical controls on chemistry of water in a cave system in Southern New Mexico. I used stable isotopes and trace elements to evaluate the relative controls of evaporation and CO₂ degassing in driving calcite precipitation in the epikarst.

Mirasol Springs Water Quality: I established 20 surface water sampling sites and monitored them monthly for two semesters. The goal of this project was to establish a baseline understanding of water quality, and evaluate groundwater flow paths, residence times, and surface-water/groundwater interactions.

Work Experience

Undergraduate Research Assistant, Texas Hydro, Bureau of Economic Geology

University of Texas at Austin, Fall 2025-Spring 2026.

Hydrologic monitoring: Tested and deployed sensors and HOBO data loggers for hydrologic monitoring across central Texas

Radon in water: Standardized methods and evaluated applications of radon as a geochemical tracer of SW/GW interaction.

Hydrogeologic mapping and subsurface characterization: Leveraged geologic mapping and downhole logging to characterize hydrogeologic systems.

Undergraduate Research Assistant, Banner Research Group, Jackson School of Geosciences, University of Texas at Austin, Fall 2022-Fall 2025.

Paleoclimate Cave Monitoring: Monthly monitoring of 4 caves in the Texas Hill Country. Collected dripwater, plate calcite, and atmospheric parameters. Analysis of major ions, and trace elements.

Urban Hydrosciences water quality monitoring. Sampling of stream water in 3 watersheds around Austin. Analysis of major ions, and trace elements

Project CRESSLE: Sampling of stream water to investigate environmental justice issues in Austin.

Field Experience

Geologic Mapping

Four days mapping deformed Cretaceous strata and evaluating water wells outside of Big Bend National Park in West Texas.

Structural Geology field trip- two days mapping Dog Canyon and Persimmon Gap in Big Bend National Park.

Geology Field Camp- 6 weeks mapping and measuring section in the Tobacco Root Mountains of southwestern Montana.

Underground

20+ cave monitoring trips to Inner Space Caverns, Natural Bridge Caverns, Cave Without a Name, and Westcave Preserve in central Texas

Pecan Springs Karst Preserve; visited undeveloped caves on the property as part of the Pecan Springs Karst Preserve Science Advisory Council

Colorado Bend State Park; assisting with establishment of an ongoing cave gas monitoring project, including access to vertical caves and working in high CO₂ environments

Awards

Outstanding Hydrogeology Field Camp Undergraduate, Summer 2025

Barry Goldwater Scholarship Nominee, Fall 2024

Udall Scholarship, \$7000, Fall 2024

Guy E. Green Endowed Presidential Scholarship, \$3000, 2024-25

The Scott and Allyson Tinker Scholarship, \$5477, Summer 2024

Zeynep Atalay Undergraduate Scholarship in Geological Sciences. \$2000, 2023-24

John C. and Marian B. Maxwell Endowed Undergraduate Scholarship in Geological Sciences. \$300, 2022

Posters

Eagle, W., Abella J., Janelle, A., Banner, J. (2025). *Reconstructing Climate Change in the Pedernales River Basin*

Texas Water Resources Network, University of Texas at Austin (8 May, 2025),

Texas Climate Conference, Rice University (11 April, 2025)

Jackson School of Geosciences Research Symposium, University of Texas at Austin (14 February, 2025)

GSA Connects, San Antonio Texas, (19 October, 2025)

Thompson, T., Quinton, P., Rygel, M., **Eagle, W.**, Keller, M., Murphy, K., & Rudlaff, J. (2025, March 29). *Measuring carbon isotopic variations and their relationship to dolomitization in the Jefferson Formation (Late Devonian).*

Student Activities

Jackson School of Geosciences Representative, UT Austin Student Government, Spring 2023-Spring 2025

Officer, Undergraduate Geological Society, Fall 2024-Fall 2025

Jackson Scholars, Spring 2023-Spring 2026

Undergraduate Honors Research Program, Fall 2024-Spring 2026

Member Undergraduate Geological Society, Fall 2022-Spring 2026

Sustainability Caucus Chair, UT Austin Student Government, Fall 2023-Spring 2024

Longhorn Legislative Aid, UT Austin Student Government, Fall 2022-Spring 2023

Relevant Coursework

Vadose Zone Hydrology, Spring 2025. Quantitatively described and modeled the flow of fluids through variably saturated porous media.

Environmental Isotope Geochemistry, Spring 2025. Analyzed and developed geochemical models to analyze water chemistry data associated with Sitting Bull Falls Cave. Learned about stable and radioactive isotope systems in the context of environmental systems.

Applied Karst Hydrogeology, Fall 2024. Analyzed well level, hydrologic, and geologic data into a scientific report and presented findings to class. Learned modern hydrogeologic data collection methods, with an emphasis on karst systems. Assisted with collection and interpretation of hydrogeologic data in the field with implications for management of water resources.

Life Through Time, Fall 2024. Developed and applied cognitive frameworks for inferring processes from complex spatiotemporal patterns. Collected, catalogued, and identified over 200 fossils in collaboration with the Non-Vertebrate Paleontology Lab.

Field Geology in the Rocky Mountains, Summer 2024. Produced and interpreted geologic maps from data collected over multiple days in the field. Collaborated to measure and describe over 100 meters of stratigraphic section for an NSF funded research project that is being presented at a regional GSA conference.

The Science of Environmental Justice, Spring 2023. Designed and wrote an NSF style proposal for a study focused on investigating environmental justice issues associated with water quality in Austin. Investigated environmental injustices associated with water quality, air quality, urban heat, erosion through community based participatory research and scientific data collection and analysis.

Technical Skills

Data Analysis: Python (Pandas, Matplotlib, Numpy), ArcGIS Pro, geochemical modeling, hydrologic modeling (Hydrus 1D), Microsoft Office, Adobe Creative Cloud, database management, data analysis and visualization

Field work: Trip planning and logistics, geologic mapping, surface water sampling, well sampling, soil sampling, tree coring, karst hydrology

Technical outdoors: caving, single rope technique (SRT), rock climbing

Lab: Clean lab procedures, aqueous geochemistry